

Program : Diploma in Architecture	
Course Code : 5188	Course Title: Working Drawing III
Semester : 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To learn the concept of detailed reinforcements in RCC columns, beams, slabs and reproduce in CAD software
- To learn the method of connections in steel beam to steel columns and reproduce them in CAD software
- To prepare a portfolio of working drawings

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic knowledge about mathematics		Engineering Mathematics	1
Basic knowledge of design of steel and concrete		Design of steel and concrete	5

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Prepare structural drawing of RCC columns, beams and slab	15	Applying
CO2	Prepare structural drawings of RCC columns and footings	6	Applying
CO3	Prepare detailed drawings of different types of RCC staircases	12	Applying
CO4	Prepare detailed drawings of steel column and beam junctions	6	Applying
	Series Test	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			2
CO2	3			3			2
CO3	3			3			2
CO4	3			3			2

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Prepare structural drawing of RCC columns, beams and slab		
M1.01	Draw the cross section at center, support and the longitudinal sections of simply supported rectangular beam, showing the reinforcement and dimensions	3	Applying
M1.02	Draw the cross section at center, support and the longitudinal section of continuous beam, showing the reinforcement and dimensions	3	Applying
M1.03	Draw the cross sections at center, support and longitudinal section of a Cantilever beam, showing the reinforcement and dimensions	3	Applying
M1.04	Draw the sectional elevation and plan showing reinforcement of a one-way slab Simply supported, continuous slab.	3	Applying
M1.05	Draw the sectional elevation for two spans and the plan at the levels of reinforcement of two-way slab (simply supported, by showing all details and dimensions)	3	Applying
CO2	Prepare structural drawings of RCC columns and footings		
M2.01	Draw the cross section of rectangular and circular columns, showing all details	3	Applying
M2.02	Draw the plan and section of columns footing, showing all details	3	Applying
	Series Test	3	
CO3	Prepare detailed drawings of different types of RCC staircases		
M3.01	Draw layout of different types of RCC stair	6	Applying
M3.02	Draw plan and section of a typical RCC stair by showing all details and dimensions)	6	Applying

CO4	Understand the drawings of Steel column footing		
M4.01	Draw the elevation and cross section of a steel beam to steel column connection	6	Applying
	Series Test	3	

Text /Reference:

T/R	Book Title/Author
T1	Balagopal&Prabhu: Building Design & Civil Engineering Drawing; Spade Engineers Pvt. Ltd
T2	Limit State of Reinforced Concrete; P C Varghese
T3	Steel structures design Allan Williams, International Code of council
R1	IS 456-2000
R2	IS 800-2007
R3	Steel tables
R4	SP 16

Online resources:

Sl.No	Website Link
1	https://www.slideshare.net/Shubham115/beam-design-and-drawing-of-rcc-structures
2	https://in.pinterest.com/slpachpute/rcc/
3	https://www.engineeringcivil.com/presentation-on-reinforcing-detailing-of-r-c-c-members.html